

Sudhanshu Jha

IIT Bhubaneswar — CGPA: 7.7/10
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Education

Indian Institute of Technology Bhubaneswar

B.Tech + M.Tech, Civil Engineering

2026*
CGPA: 7.7/10

Central Board of Secondary Education

Intermediate (12th)

2023

85%

Central Board of Secondary Education

Secondary School Certificate (10th)

2021

92%

Projects

ISROGeospatial

Computer Vision — InterIIT TechMeet 14.0

GitHub

- Developed a 3-stage computer vision pipeline using **GroundingDINO**, **SAM**, and **YOLO** for object grounding, segmentation, and detection in satellite imagery.
- Integrated **Qwen2.5B** for image captioning and Visual Question Answering (VQA), enabling multimodal image understanding.
- Built and deployed a **Flask + React.js** platform on **AWS EC2** supporting 2K×2K satellite imagery.

Structural Damage Segmentation

Deep Learning — Computer Vision

GitHub Dataset

- Developed a **9-class semantic segmentation** pipeline for reinforced-concrete structural damage using **PyTorch** and **Swin Transformer**.
- Used pixel-level annotations to classify **concrete crushing**, **delamination**, **microcracks**, **spalling**, **rebar corrosion**, **structural cracks**, and **deformation**.
- Designed a multi-scale decoder with **hierarchical feature fusion** and **skip connections** to recover spatial details and localize fine-grained damage.

ZENTRA GitHub

Full Stack Development — AI Automation

- Built a full-stack social platform using **React.js**, **Express.js**, **MongoDB**, **Socket.io**, and **Zustand** for real-time communication and content sharing.
- Integrated a **Gemini-powered AI chatbot** for content generation and automated social-media actions including posting, likes, dislikes, and deletion.
- Developed real-time communities using **Socket.io**, AI-powered assistance, debounced search, and optimized Zustand state management.

Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript

ML/CV: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Transformers, Semantic Segmentation

AI/LLM: Gemini API, Qwen, LangChain, LangGraph, VQA

Web: React.js, Next.js, Node.js, Express.js, Tailwind CSS, REST APIs, Socket.io

Databases: MongoDB, MySQL, Pinecone, Appwrite

Cloud/Tools: AWS EC2, Git, GitHub, Postman, AutoCAD

Core CS: Data Structures & Algorithms, OOP, DBMS

Positions of Responsibility

Nakshatra – Astronomy & Astrophysics Society, IIT Bhubaneswar

Sep 2023 – April 2025

Events Coordinator

- Led planning and execution of astronomy and astrophysics events, coordinating logistics, participants, and event operations.
- Served as Point of Contact for the **Blast-off** event at ASTROCHAMP 2023–24 and contributed to the organizing committee for ASTROCHAMP 2024–25.

Counselling Service Team, IIT Bhubaneswar

Jul 2024 – Apr 2025

- Mentored **13 students** and organized “Treasure Hunt” and “Freshers Got Talent” events for the 2024 batch.

Extra-Curricular Activities

Competitive Programming: Solving algorithmic challenges and improving Data Structures & Algorithms skills. LeetCode